Assignment 2: Flow Over a Cylinder

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1 Governing Equations & Setup

The unsteady, 2D incompressible flow over a cylinder is simulated by solving the transient Continuity and Navier-Stokes equations. Simulations were conducted at $Re = 272$ (Laminar) and $Re = 2902$ (Standard $k-\epsilon$ turbulence model). The SIMPLE algorithm, QUICK momentum scheme, and 2nd Order Implicit transient formulation were employed with a convergence target of 10^{-6}.

2 Flow Fields & Vortex Shedding

A block-structured grid with high refinement near the cylinder was used to capture boundary layer separation. At $Re = 272$, the flow separates behind the cylinder, forming a distinct, alternating Von Kármán vortex street in the wake. However, at $Re = 2902$ using the Standard $k-\epsilon$ model, the vortex shedding is almost completely damped out, resulting in a steady, smeared, symmetric wake. This is a known limitation of the standard $k - \epsilon$ RANS formulation, which generates excessive turbulent viscosity in wake regions, suppressing the physical instabilities.

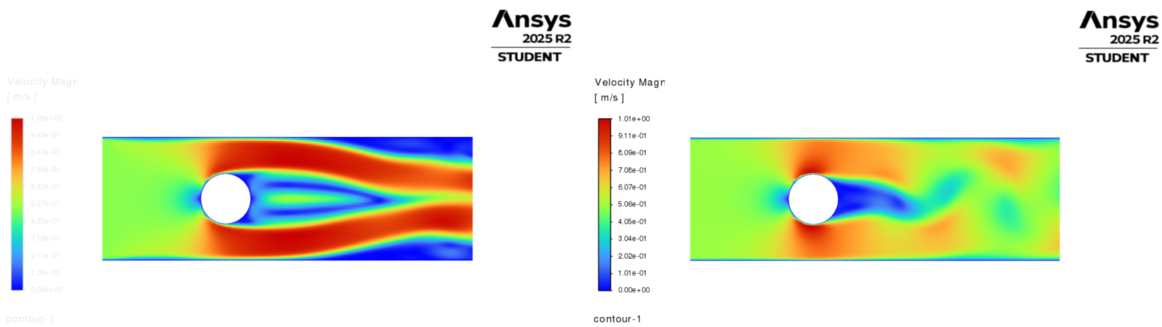

Figure 1: Velocity contours. Left: $Re = 272$ (Laminar). Right: $Re = 2902$ (Standard $k - \epsilon$).

3 Frequency Analysis

Monitoring the y-velocity in the wake over time allowed for frequency analysis. For $Re = 272$, the oscillation yielded a shedding frequency f that aligns well with the expected Strouhal number $St = fD/U \approx 0.183$. Conversely, the turbulent simulation ($Re = 2902$) showed damped probe data, failing to capture the physical shedding frequency unless more advanced models (e.g., RSM, LES) are used.